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## 📘 Tuples in C# (Complete Guide)

### 🔹 1. What is a Tuple?

A **Tuple** in C# is a lightweight data structure used to **store multiple values of different types in a single object**, without creating a class.

👉 Think of it as a **temporary container for grouped data**.

```
(string, int) person = ("John", 25);
```

Here:

- "John" → string
- 25 → int
  Both are stored together in one variable.

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### 🔹 2. Why Use Tuples?

Before tuples, we had to:

- Create a class
- Or use out parameters

👉 Tuples simplify this.

✅ **Use cases:**

- Returning multiple values from a method
- Temporary grouping of data
- Avoiding unnecessary class creation

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### 🔹 3. Tuple Types in C#

There are **two types of tuples**:

#### 1️⃣ Old Tuple (Reference Type)

```
Tuple<string, int> person = new Tuple<string, int>("John", 25);
```

Access:

```
person.Item1  // John
person.Item2  // 25
```

❌ Problems:

- Poor readability (Item1, Item2)
- Verbose syntax

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### 2️⃣ ValueTuple (Modern C#) ✅ (Recommended)

Introduced in C# 7

```
(string name, int age) person = ("John", 25);
```

Access:

```
person.name
person.age
```

✓ Cleaner
✓ Faster (value type)
✓ Readable

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### 🔹 4. Creating Tuples

**✓ Basic Tuple**

```
var person = ("John", 25);
```

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**✓ Named Tuple**

```
var person = (name: "John", age: 25);
```

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**✓ Explicit Type**

```
(string name, int age) person = ("John", 25);
```

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### 🔹 5. Accessing Tuple Values

```
var person = (name: "John", age: 25);

Console.WriteLine(person.name);
Console.WriteLine(person.age);
```

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### ✔ Without Names

```
var data = ("John", 25);

Console.WriteLine(data.Item1);
Console.WriteLine(data.Item2);
```

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### 🔹 6. Returning Multiple Values from Method

### ❌ Old Way (out parameters)

```
public void GetData(out string name, out int age)
{
    name = "John";
    age = 25;
}
```

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### ✅ Using Tuple (Best Practice)

```
public (string name, int age) GetPerson()
{
    return ("John", 25);
}
```

Usage:

```
var person = GetPerson();

Console.WriteLine(person.name);
```

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### 🔹 7. Deconstruction of Tuple

You can extract values directly into variables:

```
var person = ("John", 25);


(string name, int age) = person;


Console.WriteLine(name);

Console.WriteLine(age);
```

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**✓ Inline Deconstruction**

```
var (name, age) = ("John", 25);
```

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### 🔹 8. Discards (Ignoring Values)

```
var (name, _) = ("John", 25);


Console.WriteLine(name);  // age ignored
```

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### 🔹 9. Tuple as Method Parameters

```
public void PrintPerson((string name, int age) person)

{

   Console.WriteLine(person.name);

}
```

Call:

```
PrintPerson(("John", 25));
```

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### 🔹 10. Tuple with Collections

```
List<(string name, int age)> people = new List<(string, int)>
{
    ("John", 25),
    ("Alice", 30)
};

foreach (var p in people)
{
    Console.WriteLine(p.name);
}
```

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### 🔹 11. Nested Tuples

```
var data = (name: "John", address: (city: "Hyderabad", pin: 500001));

Console.WriteLine(data.address.city);
```

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### 🔹 12. Tuple Comparison

```
var t1 = (1, 2);
var t2 = (1, 3);

Console.WriteLine(t1 == t2); // False
```

✓ Compared element by element

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### 🔹 13. When to Use Tuple vs Class

| Scenario | Use Tuple | Use Class |
|---|---|---|
| Temporary data | ✅ | ❌ |
| Return multiple values | ✅ | ❌ |
| Complex logic/data | ❌ | ✅ |
| Reusability needed | ❌ | ✅ |

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### 🔹 14. Real-Time Example

🎯 **Scenario: Return API Response**

```
public (bool isSuccess, string message) SaveData()
{
    return (true, "Saved successfully");
}
```

Usage:

```
var result = SaveData();

if (result.isSuccess)
{
    Console.WriteLine(result.message);
}
```

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### 🔹 15. Performance Insight

- Tuple → Reference Type (Heap)

- ValueTuple → Value Type (Stack)

👉 So:
✓ ValueTuple = faster + less memory overhead

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### 🔹 16. Common Mistakes

❌ Using Tuple for complex models
❌ Not naming fields
❌ Overusing nested tuples

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### 🔹 17. Best Practices

✓ Use **ValueTuple** always
✓ Use **meaningful names**
✓ Use for **short-lived data only**
✓ Avoid deep nesting

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### 🔥 Summary

- Tuple = group multiple values
- Prefer **ValueTuple (( ))**
- Supports:
  - Naming
  - Deconstruction
  - Returning multiple values
- Not a replacement for classes

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